

Kush Ashvinbhai Patel

M.Tech – ICT (Machine Learning)

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Education

M.Tech (ICT) in Machine Learning, Dhirubhai Ambani University

CPI: 9.39/10

2025 – Present

Gandhinagar, Gujarat

B.E (CE), Madhuben & Bhanubhai Patel Institute of Technology (CVMU)

CPI: 7.81/10

2021 – 2025

Anand, Gujarat

Class 12th, Shree Vasishtha Vidhyalaya (CBSE)

Percentage: 83%

2021

Surat, Gujarat

Class 10th, Shree Vasishtha Vidhyalaya (CBSE)

Percentage: 88%

2019

Surat, Gujarat

Experience

HP Param IT Solutions

Flutter Developer Intern

Jan 2025 – May 2025

Remote

- Built and deployed a **cross-platform mobile application** using **Flutter** for an IoT-based *Street Light Automation* system, targeting Android and iOS from a single codebase.
- Collaborated with backend engineers to integrate **RESTful APIs**, enabling live device-status updates and remote actuation with seamless real-time hardware control.

Projects

Beyond Visible Spectrum: AI for Agriculture (ICPR 2026) | PyTorch, Swin Transformer, timm, Rasterio [[github](#) – [kaggle](#)]

- Placed **3rd globally in Task 2** out of **574 participants / 328 teams** in the ICPR 2026 competition, classifying crop diseases (Rust, Blast, Aphid, RPH) from **12-band Sentinel-2 satellite imagery**.
- Adapted a pretrained **Swin Transformer** (swin_tiny) for **12-channel Sentinel-2 input** via timm's channel adaptation; benchmarked against ConvNeXt, ResNet-50, and EfficientNet-B2 – Swin achieved the best result with **89.3% accuracy** and **0.78 Macro F1**.
- Applied **5-fold cross-validation** with softmax ensemble averaging and **percentile-clipped normalization** (2nd–98th) per spectral band to handle the small dataset (~900 samples) with heavy class imbalance.

AQI Vision Predictor | PyTorch, EfficientNet, CNN, OpenCV, Python [[github](#)]

- Built a system that predicts **Air Quality Index (AQI)** directly from outdoor photos instead of relying on physical sensors, turning pollution estimation into an **image classification problem**.
- Fine-tuned **EfficientNet** and trained custom **CNNs** with data augmentation and transfer learning to handle different weather, lighting, and city conditions.
- Used **OpenCV** to extract visual cues like **haze density and colour shifts** from images, which helped the model pick up on pollution-related patterns more reliably.

Earth Observation: Delhi Airshed Land-Use Classification | PyTorch, ResNet18, GeoPandas, Rasterio [[github](#)]

- Classified land use across the Delhi-NCR airshed using **Sentinel-2 satellite imagery**, filtering **8,015 usable patches** from 9,216 tiles with **GeoPandas** and **Rasterio** for spatial clipping and alignment.
- Generated **ground-truth labels automatically** by matching each patch to the **ESA WorldCover 2021** map (10m resolution), grouping 11 classes into 5 simplified land-use categories using majority voting.
- Fine-tuned **ResNet18** on the labeled patches with stratified splits, and evaluated using accuracy, **macro F1-score**, and per-class confusion matrices on held-out test data.

ShelfMind AI: Retail Shelf Intelligence | YOLO26s, DINOv2, FAISS, Streamlit, Python [\[github\]](#)

- Trained a custom **YOLO26s** object detector on the **SKU-110K** benchmark (11,762 images, 1.73M annotations) for dense product detection on retail shelves, achieving **91.7% mAP@50** at the dataset's practical annotation ceiling.
- Built a **DINOv2-based retrieval pipeline** that embeds each detected product crop into a **256-d semantic space** and matches it against a store catalog using **FAISS** nearest-neighbor search for real-time product identification.
- Developed a full **planogram compliance system** with a Streamlit dashboard that compares detected shelf layouts against expected planograms, flags misplaced or out-of-stock items, and tracks compliance trends over time.

Technical Skills

Languages: Python, C/C++, SQL, Dart, JavaScript

ML / DL Frameworks: PyTorch, Scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, HuggingFace Transformers

Web & Mobile: Flutter, Flask, FastAPI, React

Developer Tools: Git, Docker, Postman, Jupyter Notebooks, Linux Shell, VS Code

Cloud & Databases: AWS (EC2, Fargate), MongoDB, MySQL

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks

Position of Responsibility

- Core Member – Technical Wing, **AI Club, Dhirubhai Ambani University** (Aug 2025 – Present), organising workshops on Deep Learning, Computer Vision, and LLM-based agentic systems for **100+ club members**.
- Teaching Assistant – **Data Structures & Algorithms (C++)**, Dhirubhai Ambani University, mentoring 30+ undergraduate students through weekly sessions, lab evaluations, and one-on-one doubt clearing.
- Google Gemini **Student Ambassador** (Jun 2026 – Present), selected for Google's Gemini Ambassador Programme to promote Generative AI tools and conduct hands-on sessions on Gemini API and prompt engineering across campus developer communities.

Achievements

- **Amazon ML Summer School 2026:** Selected among the **top 3,000 participants** across India for Amazon's flagship ML summer programme covering deep learning, reinforcement learning, generative AI, and causal inferencing, taught by Amazon Scientists.
- [ICPR 2026 – Beyond Visible Spectrum \(3rd Place\)](#): Ranked **3rd globally in Task 2** out of **574 participants / 328 teams**, winning a **£40 Amazon prize** for synthetic Sentinel satellite data generation for crop disease classification.
- [Kaggle – Bidding Predictions for Construction \(Solo #1 / #2\)](#): Ranked **1st on Public** and **2nd on Private Leaderboard** (solo) using a multi-stage **LightGBM + XGBoost + CatBoost** ensemble with Bayesian-smoothed hierarchical pricing on 832K rows. [\[github\]](#)
- **GATE Qualified (Dual)**: Cleared GATE in **Computer Science** (Score: 456) and **Data Science & AI** (Score: 413) demonstrating strong command of core CS and AI fundamentals.
- **Competitive Programming**: [LeetCode](#) – Max Rating **1682**, 100+ solved; [CodeChef](#) – **3-Star**, Max Rating **1658**, 180+ solved; [Codeforces](#) – Max Rating **1089**, 50+ solved.
- **AI & ML Certifications**: 5-Day Gen AI Intensive (Google & Kaggle); Machine Learning (IBM); TensorFlow Developer; Deep Learning & ML Specializations (DeepLearning.AI); Structured Query Language (University of Michigan).